From the Performance-Difference Lemma to the Stochastic and Deterministic Policy-Gradient Theorems

Lecture notes for DDPG and TD3

Abstract

These notes derive two central results used in actor–critic methods: the policy-gradient theorem for stochastic policies with discrete actions, and the deterministic policy-gradient theorem for continuous actions. Both results are obtained from the performance-difference lemma.

The main conceptual point is that the performance-difference lemma compares two policies while keeping the value function of the reference policy fixed. Consequently, after taking a first-order limit, derivatives of the state occupancy measure and derivatives of the critic with respect to the policy parameters disappear. This is the mathematical reason that policy-gradient formulas have their familiar simple form.

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1 Discounted Markov decision processes

We begin by fixing notation.

1.1 The model

A discounted Markov decision process consists of:

- a state space $\mathcal{S}$;
- an action space $\mathcal{A}$;
- transition probabilities
$$P(s' \mid s, a);$$
- an expected one-step reward
$$r(s, a);$$
- an initial-state distribution ρ_0;
- a discount factor
$$0 \leq \gamma < 1.$$

The initial distribution, transition model, reward function, and discount factor are assumed not to depend on the policy parameters.

For the main proofs, we assume that the state space is finite. In the stochastic policy-gradient section, the action space is also finite. In the deterministic policy-gradient section, the action space is a subset of $\mathbb{R}^m$.

The same formulas hold for continuous state spaces after replacing sums over states by integrals and imposing the appropriate measurability, continuity, and dominated-convergence assumptions. This extension is discussed in Section 10.

We assume that rewards are uniformly bounded:

$$|r(s, a)| \leq R_{\max}.$$

This ensures that all discounted value functions are bounded. In particular,

$$|V^\pi(s)| \leq \frac{R_{\max}}{1-\gamma}, \qquad |Q^\pi(s, a)| \leq \frac{R_{\max}}{1-\gamma}.$$

1.2 Stochastic and deterministic policies

A stochastic policy π assigns an action distribution to each state:

$$\pi(a \mid s) = \mathbb{P}(A_t = a \mid S_t = s).$$

For a finite action space,

$$\pi(a \mid s) \geq 0, \qquad \sum_{a \in \mathcal{A}} \pi(a \mid s) = 1.$$

A deterministic policy μ assigns one action to each state:

$$A_t = \mu(S_t).$$

It can equivalently be regarded as a stochastic policy concentrated at one action:

$$\pi_\mu(\,\mathrm{d}a \mid s) = \delta_{\mu(s)}(\,\mathrm{d}a),$$

where δ_x denotes a Dirac probability measure at x.

1.3 Trajectories and performance

For a policy π, a trajectory is generated according to

$$S_0 \sim \rho_0, \qquad A_t \sim \pi(\cdot \mid S_t), \qquad S_{t+1} \sim P(\cdot \mid S_t, A_t).$$

The discounted return is

$$G_0 = \sum_{t=0}^{\infty} \gamma^t r(S_t, A_t).$$

The performance of π is

$$J(\pi) := \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \right]. \tag{1}$$

The expectation includes the randomness in the initial state, policy, transition dynamics, and any reward randomness. Since $r(s, a)$ denotes the expected reward, it is enough in the proofs to work with this expected one-step reward.

For a parameterized policy π_θ, we write

$$J(\theta) := J(\pi_\theta).$$

Similarly, for a parameterized deterministic policy μ_θ, we write

$$J(\theta) := J(\mu_\theta).$$

2 Value functions, advantages, and occupancy measures

2.1 Value functions

For a stochastic policy π, define its state-value function by

$$V^\pi(s) := \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \,\middle|\, S_0 = s \right]. \tag{2}$$

The state–action value function is

$$Q^\pi(s, a) := \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \,\middle|\, S_0 = s,\ A_0 = a \right]. \tag{3}$$

The Bellman equations are

$$Q^\pi(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^\pi(s'), \tag{4}$$

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) Q^\pi(s,a). \tag{5}$$

The performance can be written as

$$J(\pi) = \sum_{s \in \mathcal{S}} \rho_0(s) V^\pi(s). \tag{6}$$

For a deterministic policy μ, these equations become

$$V^\mu(s) = Q^\mu(s, \mu(s)), \tag{7}$$

$$Q^\mu(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^\mu(s'). \tag{8}$$

Notice that $Q^\mu(s,a)$ is defined for every action a, not only for $a = \mu(s)$. The first action is fixed to a, and the policy μ is followed from the next state onward.

2.2 The advantage function

The advantage of action a over the policy's usual behavior at state s is

$$A^\pi(s,a) := Q^\pi(s,a) - V^\pi(s). \tag{9}$$

For a stochastic policy, the advantage has zero expectation under that policy:

$$\begin{aligned} \sum_{a \in \mathcal{A}} \pi(a \mid s) A^\pi(s,a) &= \sum_a \pi(a \mid s) \big(Q^\pi(s,a) - V^\pi(s) \big) \\ &= \sum_a \pi(a \mid s) Q^\pi(s,a) - V^\pi(s) \sum_a \pi(a \mid s) \\ &= V^\pi(s) - V^\pi(s) \\ &= 0. \end{aligned} \tag{10}$$

For a deterministic policy μ,

$$A^\mu(s, \mu(s)) = Q^\mu(s, \mu(s)) - V^\mu(s) = 0. \tag{11}$$

These zero-advantage identities are essential in the policy-gradient derivations.

2.3 Discounted state occupancy

The unnormalized discounted state occupancy measure of π is

$$d^\pi(s) := \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\pi(S_t = s). \tag{12}$$

It is not a probability distribution, because

$$\begin{aligned} \sum_s d^\pi(s) &= \sum_{t=0}^{\infty} \gamma^t \sum_s \mathbb{P}_\pi(S_t = s) \\ &= \sum_{t=0}^{\infty} \gamma^t = \frac{1}{1-\gamma}. \end{aligned} \tag{13}$$

The normalized discounted state distribution is

$$\bar{d}^{\pi}(s) := (1-\gamma)d^{\pi}(s). \tag{14}$$

It satisfies

$$\sum_{s} \bar{d}^{\pi}(s) = 1.$$

The unnormalized convention is convenient because

$$\sum_{s\in\mathcal{S}} d^{\pi}(s)f(s) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t f(S_t)\right] \tag{15}$$

for any bounded function f.

If the normalized occupancy $\bar{d}^{\pi}$ is used instead, every policy-gradient formula below acquires an additional factor $1/(1-\gamma)$.

3 The performance-difference lemma

The performance-difference lemma gives an exact expression for the difference between the performances of two policies.

Theorem 3.1 (Performance-difference lemma)**.** *Let π and π' be two policies in the same discounted MDP. Then*

$$J(\pi') - J(\pi) = \mathbb{E}_{\pi'}\left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(S_t, A_t)\right]. \tag{16}$$

Equivalently,

$$J(\pi') - J(\pi) = \sum_{s\in\mathcal{S}} d^{\pi'}(s) \sum_{a\in\mathcal{A}} \pi'(a \mid s) A^{\pi}(s,a). \tag{17}$$

For continuous action spaces, the inner sum is replaced by an integral:

$$J(\pi') - J(\pi) = \sum_{s} d^{\pi'}(s) \int_{\mathcal{A}} A^{\pi}(s,a)\, \pi'(\,\mathrm{d}a \mid s).$$

Proof. From the Bellman equation,

$$Q^{\pi}(s,a) = r(s,a) + \gamma \mathbb{E}\left[V^{\pi}(S_{t+1}) \mid S_t = s, A_t = a\right].$$

Therefore,

$$\begin{aligned} A^{\pi}(s,a) &= Q^{\pi}(s,a) - V^{\pi}(s) \\ &= r(s,a) + \gamma \mathbb{E}\left[V^{\pi}(S_{t+1}) \mid S_t = s, A_t = a\right] - V^{\pi}(s). \end{aligned} \tag{18}$$

Suppose now that the trajectory is generated using π'. Taking the expectation of (18) at time t gives

$$\mathbb{E}_{\pi'}[A^{\pi}(S_t, A_t)] = \mathbb{E}_{\pi'}\left[r(S_t, A_t) + \gamma V^{\pi}(S_{t+1}) - V^{\pi}(S_t)\right]. \tag{19}$$

For a finite horizon T, multiply by γ^t and sum:

$$\begin{aligned} &\mathbb{E}_{\pi'}\left[\sum_{t=0}^{T} \gamma^t A^{\pi}(S_t, A_t)\right] \\ &\quad = \mathbb{E}_{\pi'}\left[\sum_{t=0}^{T} \gamma^t r(S_t, A_t)\right] + \mathbb{E}_{\pi'}\left[\sum_{t=0}^{T} \gamma^{t+1} V^{\pi}(S_{t+1}) - \sum_{t=0}^{T} \gamma^t V^{\pi}(S_t)\right]. \end{aligned} \tag{20}$$

The two sums involving V^π telescope:

$$\sum_{t=0}^{T} \gamma^{t+1} V^\pi(S_{t+1}) - \sum_{t=0}^{T} \gamma^t V^\pi(S_t) = -V^\pi(S_0) + \gamma^{T+1} V^\pi(S_{T+1}). \tag{21}$$

Consequently,

$$\mathbb{E}_{\pi'}\left[\sum_{t=0}^{T} \gamma^t A^\pi(S_t, A_t)\right] = \mathbb{E}_{\pi'}\left[\sum_{t=0}^{T} \gamma^t r(S_t, A_t)\right] - \mathbb{E}_{\pi'}[V^\pi(S_0)] + \gamma^{T+1}\mathbb{E}_{\pi'}[V^\pi(S_{T+1})]. \tag{22}$$

Because V^π is bounded and $\gamma < 1$,

$$\left|\gamma^{T+1}\mathbb{E}_{\pi'}[V^\pi(S_{T+1})]\right| \le \gamma^{T+1}\frac{R_{\max}}{1-\gamma} \longrightarrow 0.$$

Taking $T \to \infty$ in (22) gives

$$\mathbb{E}_{\pi'}\left[\sum_{t=0}^{\infty} \gamma^t A^\pi(S_t, A_t)\right] = J(\pi') - \mathbb{E}_{S_0\sim\rho_0}[V^\pi(S_0)].$$

The initial-state distribution is the same under both policies, so

$$\mathbb{E}_{S_0\sim\rho_0}[V^\pi(S_0)] = J(\pi).$$

Thus

$$J(\pi') - J(\pi) = \mathbb{E}_{\pi'}\left[\sum_{t=0}^{\infty} \gamma^t A^\pi(S_t, A_t)\right].$$

Finally, condition on S_t and use the definition of $d^{\pi'}$:

$$\begin{aligned}\mathbb{E}_{\pi'}\left[\sum_{t=0}^{\infty} \gamma^t A^\pi(S_t, A_t)\right] &= \sum_{t=0}^{\infty}\gamma^t \sum_s \mathbb{P}_{\pi'}(S_t = s)\sum_a \pi'(a \mid s)A^\pi(s,a) \\ &= \sum_s d^{\pi'}(s)\sum_a \pi'(a \mid s)A^\pi(s,a).\end{aligned} \tag{23}$$

This proves the occupancy-measure version. □

3.1 Why the occupancy belongs to the new policy

An important detail in the lemma is that

$$J(\pi') - J(\pi) = \sum_s \underbrace{d^{\pi'}(s)}_{\text{occupancy of the new policy}} \sum_a \underbrace{\pi'(a \mid s)}_{\text{new action distribution}} \underbrace{A^\pi(s,a)}_{\text{advantage of the reference policy}}.$$

Thus the occupancy and action distribution come from π', whereas the advantage comes from π. This asymmetry is what makes the result useful for differentiation: we shall choose π' to be a small perturbation of π, while keeping A^π fixed.

3.2 Deterministic specialization

Let both policies be deterministic, denoted by μ and μ'. Since μ' chooses the single action $\mu'(s)$ at state s, the performance-difference lemma becomes

$$J(\mu') - J(\mu) = \sum_s d^{\mu'}(s) A^{\mu}(s, \mu'(s)). \tag{24}$$

Using

$$A^{\mu}(s,a) = Q^{\mu}(s,a) - V^{\mu}(s)$$

and

$$V^{\mu}(s) = Q^{\mu}(s, \mu(s)),$$

we obtain the particularly useful form

$$J(\mu') - J(\mu) = \sum_s d^{\mu'}(s) \left[Q^{\mu}(s, \mu'(s)) - Q^{\mu}(s, \mu(s)) \right]. \tag{25}$$

Equation (25) will directly produce the deterministic policy-gradient theorem.

4 Continuity of the discounted occupancy

Before differentiating the performance-difference identity, we record why the occupancy measure of a perturbed policy converges to the occupancy measure of the original policy.

For a policy π, define its state-transition matrix by

$$P^{\pi}(s, s') := \sum_a \pi(a \mid s) P(s' \mid s, a). \tag{26}$$

For a deterministic policy μ,

$$P^{\mu}(s, s') := P(s' \mid s, \mu(s)). \tag{27}$$

We regard ρ_0 and d^{π} as row vectors. The discounted occupancy satisfies

$$\begin{aligned} d^{\pi} &= \sum_{t=0}^{\infty} \gamma^t \rho_0 (P^{\pi})^t \\ &= \rho_0 (I - \gamma P^{\pi})^{-1}. \end{aligned} \tag{28}$$

The inverse exists because the spectral radius of a stochastic matrix is at most one and $\gamma < 1$.

Lemma 4.1 (Continuity of discounted occupancy)**.** *Suppose $\mathcal{S}$ is finite and $P^{\pi_n} \to P^{\pi}$ entrywise. Then*

$$d^{\pi_n} \to d^{\pi}.$$

Proof. Because matrix inversion is continuous on the set of invertible matrices,

$$(I - \gamma P^{\pi_n})^{-1} \longrightarrow (I - \gamma P^{\pi})^{-1}.$$

Multiplying on the left by the fixed row vector ρ_0 gives

$$d^{\pi_n} = \rho_0 (I - \gamma P^{\pi_n})^{-1} \longrightarrow \rho_0 (I - \gamma P^{\pi})^{-1} = d^{\pi}.$$

□

For a differentiable finite-action policy π_θ, continuity of $\theta \mapsto P^{\pi_\theta}$ follows from

$$P^{\pi_\theta}(s, s') = \sum_a \pi_\theta(a \mid s) P(s' \mid s, a).$$

For a deterministic continuous-action policy μ_θ, continuity follows if both

$$\theta \mapsto \mu_\theta(s)$$

and

$$a \mapsto P(s' \mid s, a)$$

are continuous.

5 Policy gradient for stochastic policies with discrete actions

5.1 Assumptions

Let $\theta \in \mathbb{R}^p$, and let

$$\pi_\theta(a \mid s)$$

be a parameterized stochastic policy.

We make the following assumptions.

Assumption 5.1. *For every state s and action a:*

(i) $\theta \mapsto \pi_\theta(a \mid s)$ *is continuously differentiable;*

(ii) $\sum_a \pi_\theta(a \mid s) = 1$*;*

(iii) the transition model and reward function do not depend directly on θ;

(iv) the initial-state distribution ρ_0 does not depend on θ.

The dependence of the trajectory distribution on θ is entirely through the policy.

5.2 Statement of the theorem

Theorem 5.2 (Policy-gradient theorem for discrete actions)**.** *Under Assumption 5.1,*

$$\nabla_\theta J(\theta) = \sum_{s \in \mathcal{S}} d^{\pi_\theta}(s) \sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a \mid s) Q^{\pi_\theta}(s, a). \tag{29}$$

If $\pi_\theta(a \mid s) > 0$ for every relevant state–action pair, then

$$\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s) \sum_a \pi_\theta(a \mid s) \nabla_\theta \log \pi_\theta(a \mid s) Q^{\pi_\theta}(s, a). \tag{30}$$

Equivalently,

$$\nabla_\theta J(\theta) = \mathbb{E}_{\substack{S \sim \bar{d}^{\pi_\theta} \\ A \sim \pi_\theta(\cdot \mid S)}} \left[\frac{1}{1-\gamma} \nabla_\theta \log \pi_\theta(A \mid S) Q^{\pi_\theta}(S, A) \right]. \tag{31}$$

5.3 Proof using the performance-difference lemma

Proof. Fix a parameter value $\theta \in \mathbb{R}^p$. To identify the gradient, choose an arbitrary direction $u \in \mathbb{R}^p$ and consider the perturbed parameter

$$\theta_h = \theta + hu,$$

where h is a scalar approaching zero.

For brevity, write

$$\pi = \pi_\theta, \qquad \pi_h = \pi_{\theta+hu}.$$

Apply the performance-difference lemma with π as the reference policy and π_h as the new policy:

$$J(\theta + hu) - J(\theta) = \sum_s d^{\pi_h}(s) \sum_a \pi_h(a \mid s) A^\pi(s, a). \tag{32}$$

The advantage of the reference policy has zero expectation under the reference policy:

$$\sum_a \pi(a \mid s) A^\pi(s, a) = 0.$$

We may therefore subtract this zero quantity inside (32):

$$J(\theta + hu) - J(\theta) = \sum_s d^{\pi_h}(s) \sum_a \big(\pi_h(a \mid s) - \pi(a \mid s)\big) A^\pi(s, a). \tag{33}$$

Divide by h:

$$\frac{J(\theta + hu) - J(\theta)}{h} = \sum_s d^{\pi_h}(s) \sum_a \frac{\pi_{\theta+hu}(a \mid s) - \pi_\theta(a \mid s)}{h} A^{\pi_\theta}(s, a). \tag{34}$$

As $h \to 0$, differentiability of π_θ gives

$$\frac{\pi_{\theta+hu}(a \mid s) - \pi_\theta(a \mid s)}{h} \longrightarrow \langle \nabla_\theta \pi_\theta(a \mid s), u \rangle . \tag{35}$$

By Lemma 4.1,

$$d^{\pi_h}(s) \longrightarrow d^{\pi_\theta}(s). \tag{36}$$

Because the state and action spaces are finite, we may take the limit term by term. Thus

$$\begin{aligned} \lim_{h \to 0} \frac{J(\theta + hu) - J(\theta)}{h} &= \sum_s d^{\pi_\theta}(s) \sum_a \langle \nabla_\theta \pi_\theta(a \mid s), u \rangle \, A^{\pi_\theta}(s, a) \\ &= \left\langle \sum_s d^{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a \mid s) A^{\pi_\theta}(s, a), u \right\rangle . \end{aligned} \tag{37}$$

The left-hand side is the directional derivative of J in direction u:

$$DJ(\theta)[u] = \langle \nabla_\theta J(\theta), u \rangle .$$

Since (37) holds for every $u \in \mathbb{R}^p$, we conclude that

$$\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a \mid s) A^{\pi_\theta}(s, a). \tag{38}$$

We now replace the advantage by the action-value function. Since

$$A^{\pi_\theta}(s, a) = Q^{\pi_\theta}(s, a) - V^{\pi_\theta}(s),$$

we have

$$\sum_a \nabla_\theta \pi_\theta(a \mid s) A^{\pi_\theta}(s,a)$$
$$= \sum_a \nabla_\theta \pi_\theta(a \mid s) Q^{\pi_\theta}(s,a) - V^{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a \mid s). \tag{39}$$

The second term is zero because

$$\begin{aligned} \sum_a \nabla_\theta \pi_\theta(a \mid s) &= \nabla_\theta \sum_a \pi_\theta(a \mid s) \\ &= \nabla_\theta 1 \\ &= 0. \end{aligned} \tag{40}$$

Therefore,

$$\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a \mid s) Q^{\pi_\theta}(s,a),$$

which proves (29).

When $\pi_\theta(a \mid s) > 0$, the logarithmic derivative identity gives

$$\nabla_\theta \pi_\theta(a \mid s) = \pi_\theta(a \mid s) \nabla_\theta \log \pi_\theta(a \mid s). \tag{41}$$

Substituting this into (29) proves (30).

Finally, using

$$d^{\pi_\theta}(s) = \frac{\bar{d}^{\pi_\theta}(s)}{1-\gamma}$$

gives the expectation form (31). □

5.4 Baseline invariance

The action-value function can be replaced by an advantage function or by a value minus any state-dependent baseline.

Proposition 5.3 (State-dependent baselines)**.** *Let $b : \mathcal{S} \to \mathbb{R}$ be any function that does not depend on the action. Then*

$$\sum_a \nabla_\theta \pi_\theta(a \mid s) b(s) = 0. \tag{42}$$

Consequently,

$$\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a \mid s) \big(Q^{\pi_\theta}(s,a) - b(s) \big). \tag{43}$$

Proof. Since $b(s)$ does not depend on a,

$$\begin{aligned} \sum_a \nabla_\theta \pi_\theta(a \mid s) b(s) &= b(s) \nabla_\theta \sum_a \pi_\theta(a \mid s) \\ &= b(s) \nabla_\theta 1 \\ &= 0. \end{aligned} \tag{44}$$

Subtracting this zero term from the policy-gradient theorem gives (43). □

The common choice

$$b(s) = V^{\pi_\theta}(s)$$

produces

$$Q^{\pi_\theta}(s,a) - b(s) = A^{\pi_\theta}(s,a).$$

Thus the score-function form can also be written as

$$\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s) \sum_a \pi_\theta(a \mid s) \nabla_\theta \log \pi_\theta(a \mid s) A^{\pi_\theta}(s,a). \tag{45}$$

6 What happened to the derivative of the occupancy measure?

At first sight, one might expect $\nabla_\theta d^{\pi_\theta}(s)$ to appear in the policy gradient, because changing the policy changes the states that the agent visits. Nevertheless, no such term appears in Theorem 5.2.

The performance-difference proof explains why.

For a nearby policy π_h, we have

$$J(\pi_h) - J(\pi) = \sum_s d^{\pi_h}(s) \sum_a \big(\pi_h(a \mid s) - \pi(a \mid s)\big) A^\pi(s,a).$$

Informally, write the first-order expansions

$$d^{\pi_h}(s) = d^\pi(s) + O(h), \tag{46}$$

$$\pi_h(a \mid s) - \pi(a \mid s) = O(h). \tag{47}$$

Multiplying these terms gives

$$\big(d^\pi(s) + O(h)\big)O(h) = d^\pi(s)O(h) + O(h^2).$$

The change in occupancy contributes only through the product $O(h)O(h)$, which is second order. It therefore vanishes after dividing by h and taking $h \to 0$.

Equivalently, before subtracting the zero-advantage term, a derivative of the occupancy would be multiplied by

$$\sum_a \pi(a \mid s) A^\pi(s,a) = 0.$$

This is the cancellation at the heart of the policy-gradient theorem.

7 Deterministic policies with continuous actions

We now turn to the theorem used by deterministic actor–critic methods such as DDPG and TD3.

7.1 Setting and assumptions

Let the action space satisfy

$$\mathcal{A} \subseteq \mathbb{R}^m,$$

and let the policy be a differentiable function

$$\mu_\theta : \mathcal{S} \to \mathcal{A}, \qquad \theta \in \mathbb{R}^p.$$

At state s, the selected action is

$$A = \mu_\theta(s).$$

We use the following assumptions.

Assumption 7.1. *For every state s:*

(i) the map $\theta \mapsto \mu_\theta(s)$ is continuously differentiable;

(ii) the action $\mu_\theta(s)$ lies in the interior of $\mathcal{A}$, or the admissible parameter perturbations remain inside $\mathcal{A}$;

(iii) $Q^{\mu_\theta}(s,a)$ is differentiable with respect to a in a neighborhood of $a = \mu_\theta(s)$;

(iv) $a \mapsto P(s' \mid s, a)$ *is continuous for all* s, s'*;*

(v) the relevant action derivatives are bounded in a neighborhood of the current policy.

The Jacobian of the actor is denoted

$$D_\theta \mu_\theta(s) \in \mathbb{R}^{m \times p}.$$

The action gradient of the critic is the column vector

$$\nabla_a Q^{\mu_\theta}(s, a) \in \mathbb{R}^m.$$

Therefore,

$$D_\theta \mu_\theta(s)^{\mathsf{T}} \nabla_a Q^{\mu_\theta}(s, a) \in \mathbb{R}^p,$$

which has the correct dimension for a gradient with respect to θ.

7.2 Statement of the theorem

Theorem 7.2 (Deterministic policy-gradient theorem)**.** *Under Assumption 7.1,*

$$\boxed{\nabla_\theta J(\theta) = \sum_{s \in \mathcal{S}} d^{\mu_\theta}(s) D_\theta \mu_\theta(s)^{\mathsf{T}} \, \nabla_a Q^{\mu_\theta}(s, a)|_{a=\mu_\theta(s)}}. \tag{48}$$

Using the normalized discounted state distribution, this is

$$\nabla_\theta J(\theta) = \frac{1}{1-\gamma} \mathbb{E}_{S \sim \bar{d}^{\mu_\theta}} \left[D_\theta \mu_\theta(S)^{\mathsf{T}} \, \nabla_a Q^{\mu_\theta}(S, a)|_{a=\mu_\theta(S)} \right]. \tag{49}$$

In reinforcement-learning notation, the same result is often written less formally as

$$\nabla_\theta J(\theta) = \mathbb{E}_{S \sim d^{\mu_\theta}} \left[\nabla_\theta \mu_\theta(S) \nabla_a Q^{\mu_\theta}(S, a) \big|_{a=\mu_\theta(S)} \right],$$

where the convention for transposes is left implicit.

7.3 Proof using the performance-difference lemma

Proof. Fix $\theta \in \mathbb{R}^p$ and choose an arbitrary direction $u \in \mathbb{R}^p$. Define

$$\mu = \mu_\theta, \qquad \mu_h = \mu_{\theta + hu}.$$

The deterministic form of the performance-difference lemma, (25), gives

$$J(\theta + hu) - J(\theta) = \sum_s d^{\mu_h}(s) \left[Q^{\mu}(s, \mu_h(s)) - Q^{\mu}(s, \mu(s)) \right]. \tag{50}$$

It is important that Q^μ is the action-value function of the fixed reference policy $\mu = \mu_\theta$. We are not differentiating Q^{μ_h} in (50).

Divide by h:

$$\frac{J(\theta + hu) - J(\theta)}{h} = \sum_s d^{\mu_h}(s) \frac{Q^{\mu}(s, \mu_h(s)) - Q^{\mu}(s, \mu(s))}{h}. \tag{51}$$

We first evaluate the limit of the critic difference quotient. Since $\theta \mapsto \mu_\theta(s)$ is differentiable,

$$\frac{\mu_{\theta + hu}(s) - \mu_\theta(s)}{h} \longrightarrow D_\theta \mu_\theta(s) u. \tag{52}$$

Because $a \mapsto Q^\mu(s,a)$ is differentiable at $a = \mu_\theta(s)$, its first-order expansion is

$$\begin{aligned} Q^\mu(s, \mu_{\theta+hu}(s)) = Q^\mu(s, \mu_\theta(s)) \\ + \langle \nabla_a Q^\mu(s, \mu_\theta(s)), \mu_{\theta+hu}(s) - \mu_\theta(s) \rangle \\ + o(\|\mu_{\theta+hu}(s) - \mu_\theta(s)\|)\,. \end{aligned} \tag{53}$$

It follows that

$$\begin{aligned} \lim_{h\to 0} \frac{Q^\mu(s, \mu_{\theta+hu}(s)) - Q^\mu(s, \mu_\theta(s))}{h} \\ = \langle \nabla_a Q^\mu(s, \mu_\theta(s)), D_\theta \mu_\theta(s) u \rangle \\ = \left\langle D_\theta \mu_\theta(s)^\mathsf{T} \nabla_a Q^\mu(s, \mu_\theta(s)), u \right\rangle. \end{aligned} \tag{54}$$

By continuity of the deterministic policy, continuity of the transition model in the action, and Lemma 4.1,

$$d^{\mu_h}(s) \longrightarrow d^\mu(s). \tag{55}$$

Since the state space is finite, we can pass the limit through the sum in (51):

$$\begin{aligned} \lim_{h\to 0} \frac{J(\theta + hu) - J(\theta)}{h} &= \sum_s d^\mu(s) \left\langle D_\theta \mu_\theta(s)^\mathsf{T} \nabla_a Q^\mu(s, \mu_\theta(s)), u \right\rangle \\ &= \left\langle \sum_s d^\mu(s) D_\theta \mu_\theta(s)^\mathsf{T} \nabla_a Q^\mu(s, \mu_\theta(s)), u \right\rangle. \end{aligned} \tag{56}$$

The left-hand side is

$$DJ(\theta)[u] = \langle \nabla_\theta J(\theta), u \rangle\,.$$

Because u was arbitrary, the vectors inside the inner products must be equal:

$$\nabla_\theta J(\theta) = \sum_s d^{\mu_\theta}(s) D_\theta \mu_\theta(s)^\mathsf{T} \nabla_a Q^{\mu_\theta}(s, \mu_\theta(s)).$$

This is exactly (48). □

8 Interpretation of the deterministic policy gradient

The deterministic policy gradient is a composition of two derivatives.

At a given state s,

$$D_\theta \mu_\theta(s)$$

describes how changing the actor parameters changes the selected action.

Meanwhile,

$$\nabla_a Q^{\mu_\theta}(s,a)\big|_{a=\mu_\theta(s)}$$

describes how changing the action changes the predicted long-term return.

Their product,

$$D_\theta \mu_\theta(s)^\mathsf{T} \nabla_a Q^{\mu_\theta}(s, \mu_\theta(s)),$$

therefore points in a parameter direction that moves the actor's action toward actions with larger critic values.

The discounted occupancy $d^{\mu_\theta}(s)$ weights this local improvement according to how often the current policy visits state s.

8.1 No derivative of the occupancy appears

As in the stochastic case, one might expect a term involving

$$\nabla_\theta d^{\mu_\theta}(s).$$

The performance-difference proof shows why it vanishes from the first-order formula.

For a nearby policy μ_h,

$$J(\mu_h) - J(\mu) = \sum_s d^{\mu_h}(s) \left[Q^\mu(s, \mu_h(s)) - Q^\mu(s, \mu(s)) \right].$$

The critic difference is already first order:

$$Q^\mu(s, \mu_h(s)) - Q^\mu(s, \mu(s)) = O(h).$$

The occupancy change is also first order:

$$d^{\mu_h}(s) - d^\mu(s) = O(h).$$

Their product contributes only

$$O(h)\, O(h) = O(h^2),$$

which disappears in the derivative.

Another way to express the same fact is

$$A^\mu(s, \mu(s)) = 0.$$

The first-order change in occupancy is multiplied by the zero advantage of the reference action.

8.2 No total derivative of the critic appears

The theorem contains

$$\nabla_a Q^{\mu_\theta}(s, a),$$

but it does not contain an explicit parameter derivative

$$\nabla_\theta Q^{\mu_\theta}(s, a).$$

This does not mean that the value function is independent of the actor parameters. It is not. Rather, the performance-difference lemma freezes the critic of the reference policy:

$$Q^\mu(s, \mu_h(s)) - Q^\mu(s, \mu(s)).$$

Only the action argument changes in this expression. The indirect effect of future policy changes is already represented by the discounted occupancy weighting and the fixed-policy action-value function.

This cancellation is precisely what makes actor–critic methods practical: the actor can differentiate through the critic with respect to the action, without differentiating through the entire future state distribution.

9 Relation to DDPG and TD3

The deterministic policy-gradient theorem is the mathematical basis of the actor update in DDPG and TD3.

Let

- $\mu_\theta(s)$ be the actor;
- $Q_\phi(s,a)$ be a differentiable critic approximating $Q^{\mu_\theta}(s,a)$;
- $\mathcal{B}$ be a minibatch of states sampled from a replay buffer.

The exact on-policy theorem suggests the approximation

$$\widehat{\nabla_\theta J} = \frac{1}{|\mathcal{B}|} \sum_{s\in\mathcal{B}} D_\theta \mu_\theta(s)^\mathsf{T} \, \nabla_a Q_\phi(s,a)|_{a=\mu_\theta(s)} \,. \tag{57}$$

Equivalently, define the actor loss

$$L_{\text{actor}}(\theta) := -\frac{1}{|\mathcal{B}|} \sum_{s\in\mathcal{B}} Q_\phi(s, \mu_\theta(s)). \tag{58}$$

Then gradient descent on L_{actor} performs gradient ascent on the critic's estimate of return:

$$-\nabla_\theta L_{\text{actor}}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{s\in\mathcal{B}} D_\theta \mu_\theta(s)^\mathsf{T} \nabla_a Q_\phi(s, \mu_\theta(s)). \tag{59}$$

There are two approximations here:

1. The true critic Q^{μ_θ} is replaced by a learned critic Q_ϕ.
2. States are sampled from the replay-buffer distribution rather than exactly from the current discounted occupancy d^{μ_θ}.

Thus the replay-buffer actor update is generally not an exact unbiased estimate of the on-policy gradient in Theorem 7.2. It is a practical off-policy approximation.

TD3 retains the deterministic actor-gradient structure but modifies critic learning to reduce error and overestimation. Its central additions are:

1. two critics, using the smaller target value;
2. delayed actor and target-network updates;
3. target-policy smoothing.

These modifications affect how the critic and targets are constructed. The underlying actor chain rule remains the deterministic policy gradient.

10 Continuous-state formulation

When the state space is continuous or otherwise general, the discounted occupancy is a measure rather than a finite vector.

Define

$$d^\pi(B) := \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\pi(S_t \in B) \tag{60}$$

for measurable subsets $B \subseteq \mathcal{S}$.

For a measurable function f,

$$\int_{\mathcal{S}} f(s)\, d^{\pi}(\,\mathrm{d}s) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t f(S_t)\right]. \tag{61}$$

The performance-difference lemma becomes

$$J(\pi') - J(\pi) = \int_{\mathcal{S}} d^{\pi'}(\,\mathrm{d}s) \int_{\mathcal{A}} \pi'(\,\mathrm{d}a \mid s) A^{\pi}(s,a). \tag{62}$$

For a deterministic policy,

$$J(\mu') - J(\mu) = \int_{\mathcal{S}} d^{\mu'}(\,\mathrm{d}s)\left[Q^{\mu}(s,\mu'(s)) - Q^{\mu}(s,\mu(s))\right]. \tag{63}$$

Under conditions that justify passing the derivative through the integral, the stochastic policy-gradient theorem becomes

$$\nabla_\theta J(\theta) = \int_{\mathcal{S}} d^{\pi_\theta}(\,\mathrm{d}s) \int_{\mathcal{A}} \nabla_\theta \pi_\theta(a \mid s) Q^{\pi_\theta}(s,a)\ \mathrm{d}a, \tag{64}$$

when the policy has a density with respect to a fixed action-space reference measure.

The deterministic policy-gradient theorem becomes

$$\nabla_\theta J(\theta) = \int_{\mathcal{S}} d^{\mu_\theta}(\,\mathrm{d}s) D_\theta \mu_\theta(s)^{\mathsf{T}} \nabla_a Q^{\mu_\theta}(s, \mu_\theta(s)). \tag{65}$$

Sufficient regularity conditions include:

1. measurability of all policies, value functions, and derivatives;
2. continuity of the transition kernel under policy perturbations;
3. differentiability of the actor and of $Q^{\mu}(s,a)$ with respect to the action;
4. an integrable dominating function for the relevant difference quotients;
5. convergence of the perturbed discounted occupancy measures in a mode strong enough to pass to the limit.

Under these conditions, the proof is algebraically identical to the finite-state proof. Finite state spaces were used above only to avoid measure-theoretic distractions when exchanging limits, sums, and integrals.

11 Summary of the derivations

The performance-difference lemma states

$$J(\pi') - J(\pi) = \mathbb{E}_{\pi'}\left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(S_t, A_t)\right].$$

For stochastic policies with discrete actions,

$$J(\pi_{\theta+h}) - J(\pi_\theta) = \sum_s d^{\pi_{\theta+h}}(s) \sum_a \big(\pi_{\theta+h}(a \mid s) - \pi_\theta(a \mid s)\big) A^{\pi_\theta}(s,a).$$

Taking a first-order limit gives

$$\boxed{\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s) \sum_a \nabla_\theta \pi_\theta(a \mid s) Q^{\pi_\theta}(s,a)}.$$

Using the log-derivative identity,

$$\boxed{\nabla_\theta J(\theta) = \sum_s d^{\pi_\theta}(s) \sum_a \pi_\theta(a \mid s) \nabla_\theta \log \pi_\theta(a \mid s) Q^{\pi_\theta}(s,a)}.$$

For deterministic policies with continuous actions,

$$J(\mu_{\theta+h}) - J(\mu_\theta) = \sum_s d^{\mu_{\theta+h}}(s) \left[Q^{\mu_\theta}(s, \mu_{\theta+h}(s)) - Q^{\mu_\theta}(s, \mu_\theta(s)) \right].$$

Taking a first-order limit and applying the ordinary chain rule gives

$$\boxed{\nabla_\theta J(\theta) = \sum_s d^{\mu_\theta}(s) D_\theta \mu_\theta(s)^\top \nabla_a Q^{\mu_\theta}(s, \mu_\theta(s))}.$$

The most important conceptual observation is:

> The policy gradient does account for how the policy changes future states and rewards, but the performance-difference lemma organizes those effects so that neither an explicit derivative of the occupancy measure nor an explicit parameter derivative of the critic appears in the final first-order formula.